Dean M. Bisogno

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SUMMARY Data engineer and scientist who designs and builds the pipelines and platforms that turn messy operational data into reliable, production-grade systems. Pairs formal mathematical training and peer-reviewed research with hands-on experience standing up medallion lakehouse architectures, automated ETL across cloud and on-premises environments, and data pipelines serving multiple business units. Raises the technical bar on teams through rigorous problem formulation and end-to-end ownership from ingestion to insight.

PROFESSIONAL EXPERIENCE

People Analytics Director, Spring 2022 - Current.
JBS USA, Greeley, CO. Enterprise analytics and data engineering.

- Lead and mentor a three-person analytics team (a manager, a data scientist, and an analyst), having elevated a team member into the manager role, while advising on technical hiring and skill development across other business units.
- Serve as technical advisor to a senior compensation manager I recruited and trained to take over program administration, refocusing my role on analytics and data engineering.
- Help lead the company's AI modernization group: building the agentic-coding business case in collaboration with the executive team, driving adoption across business units, onboarding the team on Claude and Snowflake Cortex, and helping roll out the company's first HR chatbot agent.
- Oversee continued buildout and evolution of the enterprise data platform, including strategic migration from Dataiku to Fivetran/dbt as business needs scaled.
- Lead multi-phase migration and consolidation of fragmented data sources into an enterprise medallion lakehouse architecture serving 10 business units operating 24/7.
- Establish data governance standards across modeling, pipeline development, and reporting that are followed by teams outside the immediate analytics function.
- Design and maintain SQL and Python-based automated ETL pipelines and REST/SOAP API integrations to automate business processes and reporting across on-premises and cloud environments.
- Run a production absence-reporting pipeline I built, spanning Fivetran/Python API ingestion, Snowflake staging, dbt transformation, and Qlik reporting across Snowflake and BigQuery sources, surfacing unclassified absence events through paycode analysis and worked-hours-per-shift reconciliation of timecard data.
- Build Python analysis and internal applications on the modeled Snowflake data, including a REST API and web front end deployed for daily use by frontline teams; prototyping additional case-tracking services that reduce organizational risk exposure.
- Architect requirements, workflows, and system design for JBS's in-house HR platform supporting performance reviews, KPIs, expense reporting, vacation requests, merit and bonus administration, and employee surveys across 280,000 team members internationally.

Volunteer Data Engineer, Summer 2026 - Current.
Institut Sangalkam (Senegal education non-profit), think tank.

- Sole engineer on a three-person think tank. Build and maintain a Python pipeline that extracts national education census data from proprietary Microsoft OLAP cube files into a SQLite star schema: 25 fact tables, 53 conformed dimensions, and 2M rows covering 19,000 schools, served through an internal Streamlit application for the research team.
- Keep extraction (Windows-only ADOMD.NET driver) separate from a portable, stdlib-only build stage joined by a CSV contract, and enrich the model with World Bank, UNESCO, UN population, and geolocated school data.
- Develop a spec-driven pipeline that extracts regional education indicators from national statistics agency PDF reports, gated by validation checks against World Bank benchmarks and internal consistency before anything publishes.

Data Scientist II, Spring 2021 - Spring 2022.
JBS USA, Greeley, CO.

- Operated as an embedded data scientist focused on industrial workforce planning, independently owning problem identification, data engineering, modeling, and executive-level presentation of results.
- Co-led platform evaluation with IT, selecting Dataiku and Snowflake as core data platforms and beginning initial buildout of ETL pipelines and Python/Flask-based tooling that remain in production today.
- Rapidly designed, built, and deployed VaxTrax, an in-house COVID-19 vaccination tracking system, standing it up from scratch to meet an urgent business need ahead of OSHA emergency directives.
- Built a graph-theoretic model of viral transmission through processing facilities, incorporating masking policies, vaccination rates, and variant R0 values to forecast policy efficacy during surges.
- Deployed XGBoost models to identify high-risk employee turnover with 70% accuracy, and published an analysis linking shift length and task switching to injury rates, both informing workforce policy changes.
- Named 2022 Innovator by JBS Center for Innovation and Technological Excellence.

Graduate Research & Teaching Assistant, Autumn 2015 - Spring 2021.
Mathematics Department, Colorado State University, Fort Collins, CO.

- Built research software in Python, Magma, and Gap to automate verification of open problems in algebraic number theory, contributing to two peer-reviewed publications.
- Led a six-person undergraduate research team studying curves amenable to cryptographic applications, mentoring students in mathematical theory and scientific programming.
- Taught courses in Cryptography, Group Theory, Linear Algebra, and Calculus. Awarded Outstanding Graduate Teaching Assistant.

Lead Developer, Autumn 2009 - Spring 2015.
Paladin Innovators, Federal Way, WA.

- Built early live video streaming platforms spanning server infrastructure (C#, Python), native iOS and web clients, and interactive viewer control — filing three provisional patents on systems that anticipated capabilities

now standard in cloud-based streaming.

EDUCATION

Doctor of Philosophy, Mathematics. Winter 2020.
Dissertation: *Arithmetic Properties of Curves and Jacobians*
Colorado State University, Fort Collins, CO.

Master of Science, Mathematics. Spring 2018.
Thesis: *Abhyankar's Inertia Conjecture for Certain Sporadic Groups*
Colorado State University, Fort Collins, CO.

RESEARCH

Abhyankar's Inertia Conjecture for Some Sporadic Groups. Dean Bisogno. Preprint. arXiv:2010.01455 (2020).

Group-theoretic Johnson classes and a non-hyperelliptic curve with torsion Ceresa class. Dean Bisogno, Wanlin Li, Daniel Litt, Padmavathi Srinivasan. Épijournal de Géométrie Algébrique Volume 7 (2023). doi.org/10.46298/epiga.2023.volume7.6849.

The Supersingularity of Hurwitz Curves. Erin Dawson, Henry Frauenhoff, Michael Lynch, Amethyst Price, Seamus Somerstep, Eric Work, Dean Bisogno, Rachel Pries. Involve, a Journal of Mathematics 12-8 (2019). doi.org/10.2140/involve.2019.12.1293.

CLOUD/DATA PLATFORMS

Snowflake, dbt, Dataiku, MongoDB, QlikSense, Redis.

AI/ML

PyTorch, Scikit-learn, XGBoost, Claude Code, Snowflake Cortex, Github Copilot.

LANGUAGES & FRAMEWORKS

Python, SQL, NumPy, Pandas, Flask, C#, JavaScript, TypeScript, Node.js.

DEV OPS

Docker, Vertex, Render, Windows Server, IIS, Apache, Git, Azure DevOps.